

# Indian Railways Delay Compounding & System Architecture

Repository File Catalog, Outdated Component Audit & Compounding Delay Model Formulation

## 1. Mathematical Formulation: How Delay is Accumulated

The BlackSheep engine evaluates delay accumulation through a 5-stage sequential physical and operational pipeline. Each stage builds realistically upon actual corridor physics, dispatching priority rules, and kinetic timetable buffers.

### Stage 1: Primary En-Route Physical Shocks

**PrimaryDelay = d\_weather + d\_tsr + d\_acp + d\_engine\_failure**

- **d\_weather**: Environmental drag (Fog capped at 60 km/h: +35m, Heavy Rain: +12m, Thunderstorms: +18m, Snow: +22m).
- **d\_tsr**: Temporary Speed Restrictions (Minor track cautions 30-50 km/h: +8m, Major engineering blocks 15-20 km/h: +24m).
- **d\_acp**: Alarm Chain Pulling (+15.0m per event for brake recharge and cab reset).
- **d\_engine\_failure**: Locomotive traction motor/engine failure (+15.0m direct troubleshooting penalty).

### Stage 2: Cascading Domino Shocks Across TRETAroutenumber Zones

**CascadeDelay = d\_headway\_treta + d\_crossing + d\_platform\_hold + d\_crew\_delay**

- **Universal Block Occupancy Rule**: Each route is partitioned into sequential block elements (treta\_zone\_id). When any train ahead is delayed by  $\Delta t_{\text{ahead}}$ , the trailing train inherits that exact delay:  $\Delta t_{\text{behind}} = \Delta t_{\text{ahead}}$ .
- **Single-Track Crossing Conflicts**: Opposing trains on single-line sections incur waiting delays equal to opposing train delay plus clearance traversal time:  $\Delta t_{\text{crossing}} = \Delta t_{\text{opposing}} + T_{\text{zone\_traversal}}$ .
- **Terminal Platform Holding (d\_platform\_hold)**: Station throat saturation outer signal hold (+15.0m).
- **Crew Duty Expiry (d\_crew\_delay)**: 10-Hour statutory duty rule callout detention (+45.0m).

### Stage 3: Gross Disturbance Aggregation

**GrossDelay = PrimaryDelay + CascadeDelay**

### Stage 4: Calibrated Timetable Buffer (EA) Delay Absorption

**EA\_calibrated = max(0, Duration - (Ideal\_Running\_Time + Total\_Halts\_Time + T\_kinetic))**  
**Absorbed\_by\_EA = min(GrossDelay \* Recovery\_Rate, 0.20 \* EA\_calibrated)**

- **T\_kinetic**: Stopping/starting cycle loss (1.85-2.35 mins per halt) is subtracted from timetable slack so it is not double-counted as cascading delay.
- **Recovery Rate**: Premium T1 = 35%, Superfast T2 = 22%, Express T3 = 12% based on corridor dispatching priority.

### Stage 5: Final Destination Net Delay & Strict Lateness Threshold

**NetDelay = max(0, GrossDelay - Absorbed\_by\_EA)** **Predicted\_ETA = Scheduled\_Arrival + NetDelay**  
**Arrival\_Status = 'ON TIME' if NetDelay <= 5.0 mins else 'LATE'**

## 2. Repository Files Catalog & Applicability Audit

Below is the complete file-by-file audit of all repository components, indicating their operational work, and explicitly marking legacy or superseded files as **NOT APPLICABLE ANYMORE - OUTDATED**.

### A. Root Directory Files

| File Name                      | Status                 | Work & Role Description                                                                                                                                                  |
|--------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AppServer.py                   | ACTIVE                 | FastAPI backend server. Serves API endpoints (/api/predict, /api/search, /api/status) and static frontend. Routes all prediction requests directly through SheepBand.py. |
| SheepBand.py                   | ACTIVE                 | Core decoupled orchestration engine. Extracts scenario rows from master_pnc, fetches train's calibrated EA, and executes the calibrated EA formula-based subtraction.    |
| UploadAllToNeon.py             | ACTIVE                 | Master PostgreSQL synchronizer. Populates all 15 relational tables in Neon DB preserving foreign keys, lowercase identifiers, and constraints (1,053,052 total rows).    |
| SetupNeonRelationalSchema.py   | ACTIVE                 | PostgreSQL DDL schema architect. Enforces PRIMARY KEY, FOREIGN KEY (ON DELETE CASCADE), indexes, and data types across Neon DB.                                          |
| Orchestrar.py (root)           | NOT APPLICABLE ANYMORE | Outdated 1.5 KB stub redirecting to Pipelines. Superseded by SheepBand.py and Pipelines/Orchestrar.py.                                                                   |
| MLCompoundDelayModel.py (root) | NOT APPLICABLE ANYMORE | Outdated 376-byte stub. Superseded by Pipelines/MLCompoundDelayModel.py.                                                                                                 |
| 03SeedDb.py                    | NOT APPLICABLE ANYMORE | Outdated early SQLite/PostgreSQL seeder without relational constraints. Superseded by UploadAllToNeon.py.                                                                |
| 04SeedHaltsDb.py               | NOT APPLICABLE ANYMORE | Outdated halts table seeder without foreign keys. Superseded by UploadAllToNeon.py.                                                                                      |
| ntes_trains.db                 | NOT APPLICABLE ANYMORE | Outdated local SQLite database file. All live operational data is now centralized on Neon PostgreSQL.                                                                    |
| all_trains.json                | ACTIVE (DATA)          | Raw scraped Indian Railways NTES schedule dataset (52 MB). Source of truth for 10,620 train timetables.                                                                  |
| requirements.txt               | ACTIVE                 | Python dependencies specification (fastapi, uvicorn, sqlalchemy, psycpg2, pandas, numpy, reportlab).                                                                     |

### B. Pipelines Directory Files

| Pipeline File                            | Status | Work & Role Description                                                                                                        |
|------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------|
| Pipelines/01Preprocess.py                | ACTIVE | Calculates calibrated EA (net_slack_mins) by subtracting kinetic stopping loss (1.85-2.35m/halt) from gross schedule duration. |
| Pipelines/01bKineticAccelDecelDelay.py   | ACTIVE | Davis equation kinetic traction resistance, service braking (0.6-0.8 m/s <sup>2</sup> ), and acceleration rates.               |
| Pipelines/GenerateTRETARoutes.py         | ACTIVE | Generates TRETAroutenumber keys for 10,620 trains and creates 200,126 station-hop zone block elements.                         |
| Pipelines/GenerateMasterPnC.py           | ACTIVE | Generates Cartesian product of all 10 attributes without duplicate entries (38,880 rows) and uploads to Neon master_pnc table. |
| Pipelines/05CascadeBlockHeadway.py       | ACTIVE | Block headway cascade updated to use TRETAroutenumber zone occupancy ( $\Delta t_{\text{behind}} = \Delta t_{\text{ahead}}$ ). |
| Pipelines/06CascadeJunctionStarvation.py | ACTIVE | Models junction throat holding and outer signal starvation delays across major junctions.                                      |
| Pipelines/07CascadeTurnaroundLinkage.py  | ACTIVE | Models secondary pitline maintenance turnaround delays for rakes arriving late.                                                |
| Pipelines/08CascadeCrossingConflicts.py  | ACTIVE | Single-track crossing conflict model computing detention as opposing delay + crossing traversal time.                          |

|                                      |                        |                                                                                                                     |
|--------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------|
| Pipelines/09CascadeCrewDutyExpiry.py | ACTIVE                 | Models crew 10-hour duty hour expiry callout detentions (+45 to 75 mins).                                           |
| Pipelines/MLCompoundDelayModel.py    | ACTIVE                 | Compound delay model incorporating abnormal failures (ACP +15m, Engine Failure +15m).                               |
| Pipelines/Orchestrar.py              | ACTIVE                 | Master pipeline combining all 7 cascade delay CSVs, calibrated EA absorption, and 5-minute lateness rule.           |
| Pipelines/02GeneratePermutations.py  | NOT APPLICABLE ANYMORE | Outdated generator producing only 45 coarse permutations. Superseded by GenerateMasterPnC.py (38,880 combinations). |
| Pipelines/CalculateExtraTime.py      | NOT APPLICABLE ANYMORE | Outdated calculation of EA without kinetic halt loss subtraction. Superseded by Pipelines/01Preprocess.py.          |
| Pipelines/VerifyDelayVsExtraTime.py  | NOT APPLICABLE ANYMORE | Outdated diagnostic script used during early research on uncalibrated EA.                                           |

## C. Frontend & Core Dataset Files

| File Name                             | Status                 | Work & Role Description                                                                                                                             |
|---------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend/index.html                   | ACTIVE                 | Live web dashboard with train search, incident dropdowns (ACP, Engine failure, TRETA block delay), TRETA route badge, and safe numerical rendering. |
| Frontend/app.js                       | ACTIVE                 | Multi-train analytics UI script for KPI metrics, batch charts, and fleet summary tables.                                                            |
| Frontend/app.css                      | ACTIVE                 | CSS stylesheet for the analytics dashboard layout.                                                                                                  |
| Train-Data/masterPnC.csv              | ACTIVE                 | 38,880 unique combinations of all 10 attributes without duplicate entries (7.39 MB). Core lookup dataset.                                           |
| Train-Data/TRETARoutes.csv            | ACTIVE                 | 10,620 trains mapped to corridor TRETAroutenumber keys.                                                                                             |
| Train-Data/TRETAZoneElements.csv      | ACTIVE                 | 200,126 station-hop sequential block zone elements (treta_zone_id).                                                                                 |
| Train-Data/EACalculation.csv          | ACTIVE                 | Calibrated timetable slack buffers (net_slack_mins) after kinetic stopping loss subtraction.                                                        |
| Train-Data/FinalOrchestraOutput.csv   | ACTIVE                 | 10,620 trains master orchestrated output with calibrated EA and 5.0m lateness threshold.                                                            |
| Train-Data/PnCOutput.csv              | NOT APPLICABLE ANYMORE | Outdated 45-permutation lookup dataset (41.4 MB). Superseded by masterPnC.csv.                                                                      |
| Train-Data/MLCompoundPermutations.csv | ACTIVE                 | 19,200 multi-attribute simulation permutation test records in Neon PostgreSQL.                                                                      |

## D. Scratch & Diagnostic Scripts

| Script Name                      | Status            | Work & Role Description                                                                                                              |
|----------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| scratch/test_api_treta.py        | ACTIVE TEST       | Live automated integration test verifying API predict endpoint across TRETA blocking, ACP, and engine failure.                       |
| scratch/test_treta_scenarios.py  | ACTIVE TEST       | Direct unit test suite verifying TRETA delay inheritance ( $\Delta t_{\text{behind}} = \Delta t_{\text{ahead}}$ ) and ACP penalties. |
| scratch/upload_treta_to_neon.py  | ACTIVE MIGRATION  | Migration script that created and populated treta_routes and treta_zone_elements in Neon PostgreSQL.                                 |
| scratch/verify_fk_enforcement.py | ACTIVE VALIDATION | Integrity check validating that foreign key violations are rejected by Neon PostgreSQL.                                              |
| scratch/lowercase_cols.py        | ONE-OFF MIGRATION | One-off script that normalized mixed-case quoted SQL column names to unquoted lowercase.                                             |
| scratch/check_db.py, check_*.py  | ONE-OFF SCRATCH   | Temporary diagnostic scripts used during DB inspection. Superseded by UploadAllToNeon.py.                                            |